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SIMATS
ENGINEERING

SIMATS Online Programming Lab
DS PROGRAMMING

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QUESTIONS 0 / 5 completed

Quick Sort ▾

QUICK SORT

Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

PROGRAM EDITOR C ▾ Run Save

```
1 #include <stdio.h>
2
3 void swap(int *a, int *b) {
4     int temp = *a;
5     *a = *b;
6     *b = temp;
7 }
8
9 int partition(int arr[], int low, int high) {
10     int pivot = arr[high];
11     int i = low - 1;
12
13     for (int j = low; j < high; j++) {
14         if (arr[j] < pivot) {
15             i++;
16             swap(&arr[i], &arr[j]);
17         }
18     }
19     swap(&arr[i + 1], &arr[high]);
20     return i + 1;
21 }
```

OUTPUT

Sorted array: 9 18 23 32 50 61

INPUT

50 23 9 18 61 32

QUESTIONS 0 / 5 completed

Quick Sort

QUICK SORT

Write a program to sort the array 50, 23, 9, 18, 61, 32 using Quick Sort.

PROGRAM EDITOR

C

Run

Save

```
23 void quickSort(int arr[], int low, int high) {
24     if (low < high) {
25         int pi = partition(arr, low, high);
26         quickSort(arr, low, pi - 1);
27         quickSort(arr, pi + 1, high);
28     }
29 }
30
31 int main() {
32     int arr[] = {50, 23, 9, 18, 61, 32};
33     int n = 6;
34
35     quickSort(arr, 0, n - 1);
36
37     printf("Sorted array: ");
38     for (int i = 0; i < n; i++) {
39         printf("%d ", arr[i]);
40     }
41
42     return 0;
43 }
```

OUTPUT

```
Sorted array: 9 18 23 32 50 61
```

INPUT

50 23 9 18 61 32